

Open-Ended Evolving Organization (O Ω)

An Axiomatic, Architectural, and Falsifiable Framework for Collective Intelligence Beyond Multi-Agent Coordination and Recursive Organizational Self-Improvement

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Abstract

Artificial intelligence is increasingly organized as systems of interacting agents rather than as isolated models. Yet the existence of many agents, persistent memory, parallel execution, or even self-modifying members does not by itself constitute a highly intelligent organization. This paper proposes **Open-Ended Evolving Organization (O Ω)** as a distinct theoretical class of organizational intelligence. O Ω is defined as a persistent collective cognitive system that can preserve identity across member turnover; integrate distributed observations into evidence-bearing organizational memory; dynamically allocate roles, expertise, and communication structure; learn from recurrent outcomes; modify its own coordination and institutional mechanisms; improve the process by which such modifications are generated and evaluated; autonomously identify important unknowns; convert discoveries into shared knowledge, organizational tools, roles, protocols, and institutions; and use those changes to expand the organization's future problem-solving and discovery frontier without a predefined terminal organizational form.

The defining cycle is **distributed experience** $\rightarrow$ **evidence** $\rightarrow$ **organizational knowledge** $\rightarrow$ **institutional reorganization** $\rightarrow$ **organizational tool** $\rightarrow$ **improved collective intelligence** $\rightarrow$ **deeper frontier** $\rightarrow$ **new distributed experience**. The framework separates *individual intelligence level* from *organizational intelligence level*. An O Ω organization need not contain only I Ω individuals; heterogeneous populations of reactive, persistent, adaptive, self-improving, discovery, and open-ended individuals may coexist and perform complementary functions. Indeed, a central mechanism of advanced organization is **cognitive compression**: difficult discoveries made by high-level members are transformed into procedures, tools, and infrastructure that lower-level members can execute reliably and cheaply. Conversely, the organization must support upward flows in which local anomalies, failures, and discoveries can change shared knowledge and eventually organizational structure.

We formalize organizational state, propose twelve axioms of O Ω , distinguish O Ω from ordinary multi-agent systems, swarms, learning organizations, and civilizations, and define measurable criteria including collective synergy, expert-leverage efficiency, organizational-memory retention, knowledge-diffusion latency, cognitive-compression ratio, exploration diversity, institutional improvement competence, topology-adaptation gain, turnover resilience, false-knowledge propagation, and organizational frontier expansion. A falsifiable experimental program is proposed. Contemporary work on collective intelligence, organizational learning, organizational memory, open-ended multi-agent evolution, group-evolving agents, autonomous multi-agent discovery, and institution design provides important components, but no current system demonstrates the complete O Ω property set. The central thesis is that the highest organizational intelligence is not the organization with the smartest member or the largest number of agents, but the organization that can continually transform distributed experience into collective knowledge, collective knowledge into better organizational machinery, and better organizational machinery into access to previously unreachable problems and discoveries while preserving evidence, continuity, and governance.

Keywords: organizational intelligence; collective intelligence; multi-agent systems; open-ended evolution; organizational learning; recursive self-improvement; artificial organizations; collective memory; AI agents; autonomous discovery

1. Introduction

The recent development of language-model agents has shifted attention from isolated model capability toward systems of interacting agents. Modern agent systems may include planners, coders, researchers, critics, verifiers, memory services, routers, managers, and tool-using specialists. Multiple agents can run asynchronously, maintain separate memories, exchange results, share persistent artifacts, or evolve in parallel. This makes a question increasingly important: **what makes an organization itself intelligent?**

The question cannot be answered by counting agents. Ten agents connected by a chat loop may perform worse than one expert. A team may fail to exploit the knowledge of its strongest member. Shared memory may amplify errors rather than knowledge. Consensus can destroy useful diversity. More communication can increase cost without increasing insight. Conversely, a group of moderately capable members can outperform stronger individuals if the organization preserves complementary expertise, routes work well, retains knowledge, and learns from repeated outcomes.

Human research already motivates this distinction. Woolley et al. reported evidence for a collective-intelligence factor in small groups that was not reducible to the maximum individual intelligence in the group [1], although later work challenged how independent such a factor is from individual ability. Organizational-learning theory has long distinguished changing actions from changing the governing assumptions behind actions [2,3]. March showed that organizations must balance **exploration** of new possibilities against **exploitation** of established knowledge [4]. Walsh and Ungson treated organizational memory as a distinct problem of acquisition, retention, and retrieval [5]. Nonaka emphasized that organizations can amplify knowledge created by individuals through structured knowledge-conversion processes [6].

The same problems are now appearing in artificial multi-agent systems. Recent studies show both opportunity and failure. Group-Evolving Agents use experience sharing across evolving agents to improve long-term progress [7]. CORAL uses long-running, persistent, asynchronous agents and shared memory for open-ended discovery [8]. Work on institutional design suggests that governance topology can strongly change multi-agent performance and may itself need to evolve as agents and tasks evolve [9]. At the same time, recent experiments show that unconstrained LLM teams can systematically underuse their expert member and converge toward inferior compromise [10]. Thus, **more agents are not automatically more intelligence**.

This paper proposes a theoretical endpoint for organizational development: **O Ω — Open-Ended Evolving Organization**. It is intended as the organizational counterpart to I Ω — Open-Ended Self-Evolving Individual Intelligence. The two concepts are related but not reducible to one another. An individual may become open-ended while belonging to a primitive organization; an organization may display sophisticated collective adaptation even if most members are individually modest. The correct description of an artificial society, laboratory, company, or research collective therefore requires at least two coordinates: one for individual developmental depth and one for organizational developmental depth.

The paper makes six contributions.

First, it defines the organizational unit of analysis and separates organizational identity from the identities of its members.

Second, it introduces a developmental hierarchy from coordination through persistence, adaptation, organizational self-improvement, recursive institutional self-improvement, autonomous collective discovery, and finally open-ended organizational evolution.

Third, it proposes twelve axioms of O Ω , emphasizing organizational memory, heterogeneous intelligence, evidence-bearing communication, dynamic specialization, exploration diversity, institutional plasticity, recursive meta-learning, turnover resilience, and frontier expansion.

Fourth, it explains how all individual intelligence levels can coexist inside one advanced organization and introduces **cognitive compression** as the downward transformation of high-level cognition into lower-cost procedures and infrastructure.

Fifth, it gives a candidate architecture and measurable criteria for O Ω .

Sixth, it proposes a falsifiable experimental program. O Ω should not be assigned because a system looks complex or calls itself self-organizing. It should be earned by longitudinal evidence that organizational changes improve future organizational learning, discovery, and reconfiguration under changing conditions.

2. The Unit of Analysis: What Is an Organization?

2.1 An organization is not merely a set of agents

Let an organization at time or generation t be

$$\mathbf{O}_t = (\mathbf{A}_t, \mathbf{K}^\wedge \mathbf{O}_t, \mathbf{U}^\wedge \mathbf{O}_t, \mathbf{M}^\wedge \mathbf{O}_t, \mathbf{G}^\wedge \mathbf{O}_t, \mathbf{R}_t, \mathbf{N}_t, \mathbf{P}^\wedge \mathbf{O}_t, \mathbf{\Theta}^\wedge \mathbf{O}_t, \mathbf{\Gamma}_t, \mathbf{C}_t).$$

The components are:

$\mathbf{A}$ — the current population of individual members, each with its own intelligence level and state;

$\mathbf{K}^\wedge \mathbf{O}$ — organizationally accepted knowledge and models;

$\mathbf{U}^\wedge \mathbf{O}$ — represented organizational unknowns, contradictions, capability gaps, and open questions;

$\mathbf{M}^\wedge \mathbf{O}$ — organizational memory independent of any one member;

$\mathbf{G}^\wedge \mathbf{O}$ — shared goals, missions, portfolios, and collective intentions;

$\mathbf{R}$ — roles, responsibilities, expertise assignments, and task allocation;

$\mathbf{N}$ — communication, dependency, authority, and information-flow topology;

$\mathbf{P}^\wedge \mathbf{O}$ — shared procedures, tools, protocols, workflows, and institutional routines;

$\mathbf{\Theta}^\wedge \mathbf{O}$ — organizational cognitive mechanisms: routing, aggregation, knowledge promotion, specialization, evaluation, learning, reconfiguration, and organizational improvement;

$\mathbf{\Gamma}$ — governance and authority constraints;

$\mathbf{C}$ — collective resources: compute, time, data, instruments, money, access, and attention.

An organization forms a persistent lineage

$$\mathbf{O}_0 \rightarrow \mathbf{O}_1 \rightarrow \mathbf{O}_2 \rightarrow \dots$$

even though its membership may change.

2.2 Organizational identity survives member turnover

The organization is not identical to the current set of members. A university remains recognizable after students and faculty change. A laboratory can preserve methods, datasets, institutional memory, and scientific goals after personnel turnover. Likewise, an artificial organization must preserve enough of its memory, governance, identity, knowledge, and procedures to remain causally continuous across membership changes.

Thus an organization is persistent when:

$$\mathbf{A}_{(t+1)} \neq \mathbf{A}_t$$

can occur while

$$\mathbf{O}_{(t+1)}$$

remains a legitimate successor of $\mathbf{O}_t$.

This property becomes central at high levels of organizational intelligence because it separates organizational learning from the private learning of particular members.

2.3 Organizational intelligence is not the sum of individual intelligence

A naive model would define collective capability as

$$\text{Capability}(\mathbf{O}) = \Sigma \text{Capability}(\mathbf{I}_i).$$

This is inadequate because performance depends on communication structure, role allocation, redundancy, expertise utilization, knowledge integration, error correction, and resource competition.

A more useful abstraction is

$$\text{Capability}(\mathbf{O}) = F(\{\mathbf{I}_i\}, \mathbf{N}, \mathbf{R}, \mathbf{M}^\wedge \mathbf{O}, \mathbf{P}^\wedge \mathbf{O}, \mathbf{\Theta}^\wedge \mathbf{O}, \mathbf{\Gamma}, \mathbf{C}).$$

Two organizations with identical members can therefore have very different intelligence.

2.4 Organizational intelligence can be positive or negative

The collective can create **synergy** when organization makes members jointly better than they would be separately. It can also create **anti-synergy** when coordination degrades the best available answer. Recent LLM-team results demonstrate this possibility directly: teams can identify an expert yet still dilute the expert’s answer through integrative compromise [10].

Therefore organizational intelligence must be measured against meaningful baselines, including the best member, not merely against a random or average member.

3. Organizational Developmental Depth

This paper proposes the following progression:

O0 — Coordinated Organization: members are routed and combined; the mutable object is temporary coordination.

O1 — Persistent Organization: the organization remembers; the mutable object is organizational memory and state.

O2 — Adaptive Organization: experience changes future allocation and procedures; mutable objects include roles, routing, and shared knowledge.

O3 — Self-Improving Organization: the organization improves how it coordinates and learns; the mutable object is organizational mechanism $\Theta^A O$.

O4 — Recursively Self-Improving Organization: the organization improves how it improves; the mutable object is the organizational meta-improvement process.

O5 — Autonomous Discovery Organization: the organization collectively identifies and investigates important unknowns; the mutable object is the organizational knowledge frontier.

$O\Omega$ — Open-Ended Evolving Organization: discoveries create new organizational forms and tools that expand future collective capability; the mutable object is the organizational possibility space itself.

The scale measures **developmental depth**, not raw task performance. An O1 organization using extremely powerful models may outperform an experimental O3 organization while still belonging to a lower developmental class.

3.1 O0 — Coordinated Organization

At O0, multiple agents can divide work, exchange messages, and aggregate results, but there is little persistent organizational state. Coordination may be defined by a fixed graph or one-shot workflow.

The organization can answer:

“Who should do this task?”

but not necessarily:

“What did our organization learn from similar tasks last month?”

3.2 O1 — Persistent Organization

At O1, collective state survives individual sessions. The organization retains decisions, task history, shared artifacts, expertise maps, and institutional commitments.

$M^A O_{(t+1)} \neq M^A O_t$.

This is organizational persistence, not yet organizational learning.

3.3 O2 — Adaptive Organization

At O2, historical outcomes modify future organizational behavior. The system can learn, for example, that a member is reliable on one class of tasks, that a certain workflow fails under a particular condition, or that a type of evidence should be independently replicated.

Experience changes future routing, knowledge, role assignment, or procedure:

Experience $\rightarrow \Delta\{K^{\wedge}O, R, P^{\wedge}O, N\}$.

The adaptation rule itself may remain fixed.

3.4 O3 — Self-Improving Organization

At O3, the organization can modify its own coordination and learning machinery. It may redesign communication topology, memory promotion, role formation, verification pathways, or resource allocation because evidence shows the current mechanism is inadequate.

$\Theta^{\wedge}O_{(t+1)} \neq \Theta^{\wedge}O_t$.

This is the organizational analogue of cognitive self-improvement.

3.5 O4 — Recursively Self-Improving Organization

At O4, the process that generates and evaluates organizational improvements becomes itself improvable.

Let $A^{\wedge}O_t$ be the organizational-improvement process. O4 requires the possibility of

$A^{\wedge}O_{(t+1)} \neq A^{\wedge}O_t$

based on evidence that the new meta-process generates better future organizational changes.

This corresponds to learning how to learn at the organizational level and is related to the idea of deuterio-learning or meta-learning in organizational-learning theory [2,3].

3.6 O5 — Autonomous Discovery Organization

At O5, the organization can do more than execute assigned work. It can detect unresolved problems, generate research agendas, coordinate theory and experiment, validate new knowledge, and integrate discoveries into the shared epistemic state.

The organization can create its own epistemic goals:

$U^{\wedge}O_t \rightarrow Q^{\wedge}O_t \rightarrow H^{\wedge}O_t \rightarrow E^{\wedge}O_t \rightarrow K^{\wedge}O_{(t+1)}$.

3.7 OΩ — Open-Ended Evolving Organization

OΩ is reached only when organizational discovery, institutional change, member development, and future frontier expansion close into a recurring loop.

A discovery does not merely enter memory. It can create:

new roles;

new specialist types;

new communication structures;

new tools;

new evaluation procedures;

new forms of memory;

new ways to allocate resources;

new ways to generate organizations;

new research domains that the previous organization could not effectively pursue.

The defining requirement is not endless organizational churn. It is the demonstrated ability to **expand the organization's reachable frontier by reorganizing itself in response to validated knowledge.**

4. Definition of O Ω

4.1 Core definition

Open-Ended Evolving Organization (O Ω) is a persistent collective cognitive system composed of multiple potentially heterogeneous individual intelligences that can preserve organizational identity across member turnover; maintain evidence-bearing shared memory; integrate local observations into validated organizational knowledge; dynamically specialize, allocate, and reconnect members; modify its own coordination, learning, and governance-compatible institutional mechanisms; improve the mechanism that generates and evaluates future organizational changes; autonomously formulate and investigate significant unknowns; transform discoveries into new organizational tools, roles, protocols, and institutions; and use those changes to expand its future collective learning, problem-solving, and discovery frontier without a predefined terminal organizational form.

4.2 The defining recursive loop

The O Ω cycle can be represented as:

Distributed Experience_t $\rightarrow$ Evidence_t $\rightarrow$ K^{^O}_(t+1) $\rightarrow$ {P^{^O}_(t+1), N_(t+1), R_(t+1), Θ ^{^O}_(t+1)} $\rightarrow$ COG^{^O}_(t+1) $\rightarrow$ F^{^O}_(t+1) $\rightarrow$ New Distributed Experience_(t+1).

where:

K^{^O} is organizational knowledge;

P^{^O} is organizational procedure/tooling;

N is communication and dependency topology;

R is role and specialization structure;

Θ ^{^O} is organizational cognitive machinery;

COG^{^O} is collective cognitive capability;

F^{^O} is the organizationally reachable frontier.

A strong O Ω episode includes:

F^{^O}_(t+1) $\supset$ F^{^O}_t.

The organization becomes able to investigate or solve a class of problems that its earlier form could not reliably address under comparable resources.

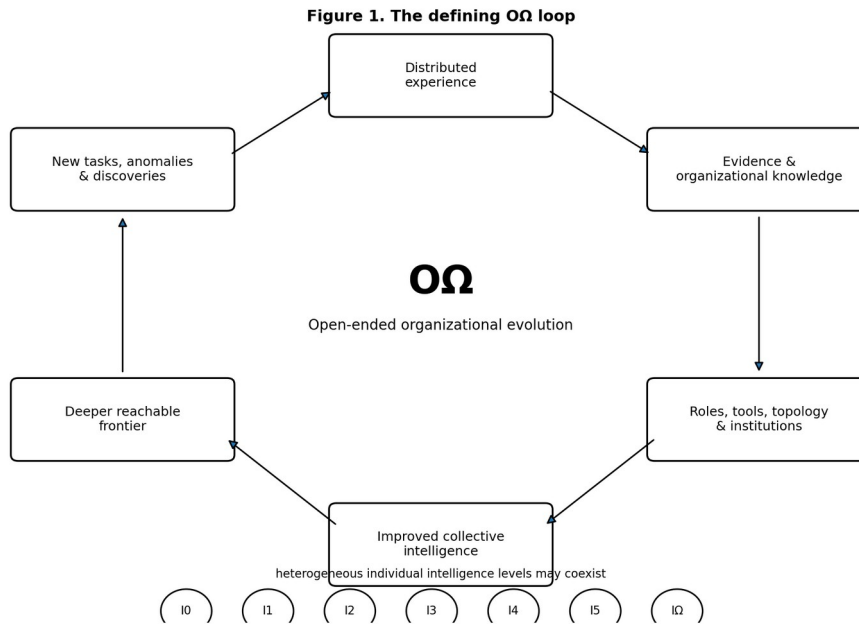


Figure 1. The defining OΩ loop. Heterogeneous individual intelligence levels can coexist while organizational experience is converted into knowledge, institutions, improved collective intelligence, and a deeper frontier.

4.3 Open-ended does not mean ungoverned

Open-ended organizational evolution concerns capability and structure, not unlimited authority. An organization may evolve its memory, topology, procedures, and internal roles while constitutional constraints, external ownership, evidence requirements, and hard safety boundaries remain independently governed.

This distinction prevents “the organization can improve itself” from becoming “the organization may redefine what counts as success or what it is allowed to do.”

5. Twelve Axioms of OΩ

Axiom 1 — Persistent Organizational Identity

OΩ must remain a causally continuous organization across time even when members, models, tools, or machines change. The lineage must retain organizational memory, provenance, commitments, and a traceable history of structural transformations.

A temporary swarm that dissolves after one task is not OΩ.

Axiom 2 — Heterogeneous Cognitive Ecology

OΩ must permit members with different intelligence levels, specialties, costs, reliabilities, and representations to coexist.

The organization does not require every member to be maximally capable. Diversity is often functional: cheap reactive members execute mature procedures; persistent members maintain local continuity; adaptive specialists learn domains; self-improving members refine methods; discovery members explore frontiers; and open-ended individuals may create new cognitive forms.

Homogeneity can simplify coordination but increases correlated failure and epistemic monoculture.

Axiom 3 — Organizational Memory Beyond Individual Memory

The organization must retain knowledge that survives the departure or failure of particular members.

Organizational memory should preserve at least:

decisions and their rationale;
evidence and provenance;
failed approaches;
expertise and reliability histories;
procedures and tools;
unresolved contradictions;
organizational changes and their measured effects;
current commitments and dependencies.

The organization must support acquisition, retention, and retrieval rather than treating a shared chat transcript as sufficient memory [5].

Axiom 4 — Evidence-Bearing Communication

Messages between members are not automatically organizational truth. Important claims travel with provenance, evidence, uncertainty, scope, and source identity.

Communication therefore has two layers:

content — what is being claimed;

epistemic metadata — why the organization should trust, question, replicate, or ignore it.

This is necessary because an advanced organization can spread errors faster than a single individual.

Axiom 5 — Local-to-Global Epistemic Promotion

Private belief does not become shared knowledge merely because the member is powerful.

For member i :

$K_i \not\Rightarrow K^O$.

Instead:

$\text{Claim}_i + \text{Evidence} + \text{Appropriate Verification} \rightarrow \text{Candidate } K^O \rightarrow \text{Promoted } K^O$.

Promotion may depend on domain-specific proof, replication, deterministic checks, independent review, or calibrated evidence aggregation.

Axiom 6 — Dynamic Specialization and Expert Utilization

The organization must be able to discover who or what is best suited for a task and adjust roles accordingly.

This includes:

forming new specialist roles;
retiring obsolete roles;
routing work by measured competence rather than title alone;
preserving generalists that connect domains;
preventing consensus from diluting expert evidence.

Recent findings that LLM teams can identify experts yet fail to leverage them make this axiom empirically nontrivial [10].

Axiom 7 — Exploration-Exploitation Balance

O Ω must preserve enough exploratory diversity to discover new possibilities while exploiting validated procedures efficiently.

March's classic result is directly relevant: adaptive systems can over-optimize exploitation and become effective in the short run while undermining long-run exploration [4].

O Ω therefore requires explicit mechanisms for:

- protected exploratory branches;
- novelty budgets;
- alternative hypotheses;
- diversity of methods;
- delayed selection where premature convergence would be harmful.

Axiom 8 — Institutional Plasticity

The organization must be able to change its own structure when evidence shows that the current structure is inadequate.

Potentially mutable organizational objects include:

- reporting lines;
- communication topology;
- role boundaries;
- memory-access rules;
- aggregation procedures;
- reviewer/executor separation;
- resource allocation;
- task decomposition;
- escalation paths;
- incentives and evaluation procedures.

This is the organizational equivalent of representation or cognitive-tool change in an individual.

Axiom 9 — Knowledge-to-Organizational-Tool Transformation

Validated discoveries must be able to become durable shared capability.

Examples include:

- a new workflow;
- a new shared tool;
- an executable procedure;
- a new role;
- a new benchmark;
- a new communication protocol;
- a new memory index;
- a new institutional rule;
- a new specialist agent template.

The defining test is whether future members can benefit from the discovery without independently rediscovering it.

Axiom 10 — Recursive Organizational Meta-Learning

The organization's method of improving itself must itself be an object of learning.

It should be able to ask:

Which kinds of organizational experiments produce useful changes?

Which evaluation methods overfit historical tasks?

Which institutional proposals generate hidden costs?

Which changes transfer across domains?

Which structures preserve expertise and diversity?

OΩ therefore requires improvement not only of the organizational mechanism Θ_O , but also of the improvement process A_O that generates and evaluates changes to Θ_O .

Axiom 11 — Turnover and Perturbation Resilience

Organizational intelligence must not disappear when one member leaves, one model is replaced, or one machine fails.

A mature organization should degrade gracefully, reconstruct roles when necessary, detect missing knowledge, and preserve critical commitments through shared memory and redundancy.

An organization that is intelligent only because one hidden expert carries everything is not organizationally intelligent in the strong sense.

Axiom 12 — Open-Ended Frontier Expansion

OΩ must be able to produce organizational transformations that make new classes of problems discoverable or solvable.

Let $F^{\wedge}O_t$ be the organizationally reachable frontier under a resource budget. OΩ requires repeated evidence of episodes where:

$$F^{\wedge}O_{(t+1)} \supset F^{\wedge}O_t$$

because the organization created or adopted new knowledge, roles, tools, institutions, or collective representations.

No fixed organization chart, fixed role taxonomy, or fixed task distribution defines the terminal form.

6. Coexistence of Individual Intelligence Levels Inside OΩ

6.1 OΩ does not require all members to be IΩ

A critical property of this framework is **orthogonality** between individual and organizational level.

An OΩ organization can, in principle, contain mostly I1–I3 members if organizational memory, institutional evolution, population diversity, and discovery mechanisms together support open-ended collective development. Conversely, a group composed entirely of highly capable I5 or IΩ individuals may remain organizationally primitive if it lacks shared memory, role allocation, evidence integration, or institutional learning.

Thus:

$\max(I_i)$ does not determine O-level.

6.2 Functional role of each individual level

I0 Reactive: deterministic or cheap execution, transformation, filtering, and formatting.

I1 Persistent: local task continuity, long-lived operational roles, and project memory.

I2 Adaptive: domain specialists that learn from repeated experience.

I3 Self-improving: local method improvement and tool/workflow redesign.

I4 Recursive self-improving: meta-improvement, cognitive-architecture research, and improvement-process design.

I5 Autonomous discovery: frontier science, problem formation, hypothesis generation, and experiment design.

IΩ Open-ended individual: invention of new cognitive representations and new classes of discovery capability.

An efficient OΩ organization routes work to the **lowest sufficient intelligence level** consistent with correctness and risk. This preserves expensive frontier cognition for the problems that actually require it.

6.3 Upward flow: local evidence can change the organization

Information should move upward through levels:

I0 anomaly → I1 episode → I2 pattern → I3 method failure → I4 improvement question → I5 research problem → I Ω new representation or capability.

A low-level member can therefore trigger a high-level organizational change if the evidence is important enough.

6.4 Downward flow: difficult cognition becomes infrastructure

High-level insight should move downward:

I Ω /I5 discovery → I4/I3 method → I2 learned workflow → I1 persistent procedure → I0 executable tool.

This creates one of the most important mechanisms of organizational intelligence: **cognitive compression**.

7. Cognitive Compression

7.1 Definition

Cognitive compression is the organizational process by which expensive, high-level cognition is transformed into reliable lower-cost representations, procedures, tools, or infrastructure that can be executed by less capable members.

A discovery may initially require extensive reasoning by an I5 scientist. Once understood, it can become an algorithm. Once the algorithm is validated, an I0 executor can run it millions of times.

7.2 Why lower-level intelligence persists

Advanced organizations should not eliminate lower levels. They create more of them.

A successful high-level discovery often reduces the intelligence required for future execution:

hard problem → discovery → abstraction → procedure → automation.

The result is an organizational pyramid in which a small amount of frontier cognition generates large volumes of inexpensive reliable execution.

7.3 Compression must preserve provenance and validity scope

Compression can also be dangerous. A rule that works under one set of conditions may be overgeneralized when turned into infrastructure.

Therefore each compressed procedure should preserve:

origin and evidence;

scope of validity;

assumptions;

known failure cases;

version;

revalidation trigger.

Cognitive compression without provenance becomes institutionalized error.

8. Organizational Memory and Knowledge Creation

8.1 Shared memory is not a giant transcript

A scalable organization cannot inject all historical communication into every member. Organizational memory should instead separate:

raw evidence and immutable records;
episodic organizational history;
validated shared knowledge;
procedures and tools;
role/expertise models;
current intentions and commitments;
institutional lineage.

This parallels the distinction between individual memory types while introducing organization-specific promotion and access rules.

8.2 Knowledge is created locally and amplified organizationally

Nonaka argued that organizations play a critical role in articulating and amplifying knowledge developed by individuals [6]. Artificial organizations make this mechanism explicit: a member can generate a discovery, but the organization must capture, validate, translate, distribute, and operationalize it.

A useful cycle is:

local observation → **explicit claim** → **shared evidence** → **validation** → **organizational abstraction** → **tool/procedure** → **distributed adoption**.

8.3 Organizational forgetting can be useful

Perfect memory is not automatically intelligent. Organizations also need mechanisms for:

superseding obsolete rules;
archiving low-value detail;
detecting stale knowledge;
preserving contradictory evidence without injecting it everywhere;
distinguishing historical context from current policy.

Forgetting at the active layer can coexist with preservation in the evidence archive.

9. Organizational Self-Model and Unknown Model

9.1 Organizational self-model

An O Ω candidate should maintain an evidence-grounded model of itself, including:

active members and roles;
measured competencies and uncertainty;
communication topology;
resource use;
bottlenecks;
memory health;
known coordination failures;
dependence on critical members;
current governance constraints;
exploration diversity;
unresolved organizational contradictions;
improvement history.

The self-model must distinguish **unmeasured** from **poor**. Absence of evidence should not become a fabricated competence score.

9.2 Organizational unknowns

O Ω should also represent what the organization does not know about itself or the world:

missing expertise;

unexplained performance gaps;

unresolved scientific questions;

inaccessible data;

unknown dependencies;

disagreement among members;

suspicious correlations;

untested institutional assumptions.

This turns organizational ignorance into a research agenda rather than a hidden failure mode.

10. A Minimal Architecture for an O Ω Candidate

A candidate implementation can be modular. The following modules are functional requirements, not claims that one software stack is uniquely correct.

10.1 Member layer

A heterogeneous population of persistent or transient agents with private state, explicit roles, measured competencies, and bounded authority.

10.2 Organizational memory layer

A shared evidence ledger, episodic history, validated knowledge store, procedure/tool registry, role/expertise model, and institutional lineage.

10.3 Communication and routing layer

Task routing, evidence-aware message transport, dependency management, broadcast limits, and topology control.

10.4 Organizational epistemic layer

Claim promotion, conflict tracking, replication requests, uncertainty propagation, source weighting, and knowledge supersession.

10.5 Organizational self-model

A continually updated representation of collective capability, member reliability, resource state, bottlenecks, diversity, and failure modes.

10.6 Exploration portfolio

A protected mechanism for maintaining multiple hypotheses, exploratory branches, and research directions rather than collapsing prematurely to one consensus.

10.7 Institution foundry

A mechanism that proposes candidate changes to roles, topology, memory policy, workflows, reviewer/executor separation, or resource allocation.

10.8 Organizational evaluator

A layer that tests candidate organizational changes against frozen historical cases, generated challenges, live canaries, cost constraints, and held-out tasks.

10.9 Organizational promoter

A distinct mechanism that activates a candidate configuration only after predeclared evaluation conditions pass. The proposer should not unilaterally become the judge of its own institutional proposal.

10.10 Cognitive-compression compiler

A process that transforms validated high-level discoveries into shared tools, lower-level procedures, agent templates, or infrastructure while preserving scope and provenance.

11. Organizational Improvement as an Experimental Process

11.1 Single-loop, double-loop, and meta-loop organization

Organizational-learning theory provides a useful analogy.

Single-loop: change actions while keeping governing organizational assumptions fixed.

Double-loop: change the assumptions or governing structures that generate actions [2,3].

Meta-loop / deuterio-learning: improve the process by which the organization learns and revises those governing structures.

These map approximately to:

O2 → O3 → O4.

11.2 Candidate institutional changes should be versioned

A proposed change to organizational topology or process should be treated like an experiment:

O_t → candidate O'_t → evaluation → promote / reject / revise.

The organization should record:

the proposed change;

causal rationale;

evidence that motivated it;

expected outcomes;

evaluation conditions;

observed results;

regressions;

rollback decision.

11.3 Local improvement can create global harm

A workflow may improve one subteam while harming information flow elsewhere. A faster routing policy may starve exploratory agents. A stronger consensus mechanism may reduce error on routine tasks while suppressing expert disagreement.

Therefore organizational promotion should use disaggregated metrics rather than one blended score.

12. Operational Criteria and Metrics

No single scalar should define O Ω . The following metrics capture distinct properties.

12.1 Collective Synergy Gain (CSG)

Measure organizational performance relative to the strongest relevant member:

CSG = Performance(O) – max_i Performance(I_i).

Positive CSG indicates genuine collective gain; negative CSG reveals anti-synergy.

12.2 Expert-Leverage Efficiency (ELE)

Measure how effectively the organization identifies and appropriately weights expertise when it exists.

A high-performing organization should not average away a demonstrably superior expert merely to preserve consensus.

12.3 Organizational Memory Retention (OMR)

Measure how much critical organizational knowledge remains usable after time, context change, and member turnover.

12.4 Knowledge-Diffusion Latency (KDL)

Measure the time or cost from validated local discovery to appropriate organizational availability.

Too slow produces repeated rediscovery. Too fast without validation produces misinformation cascades.

12.5 Cognitive Compression Ratio (CCR)

Measure how much expensive frontier cognition can be converted into lower-cost reliable execution.

A conceptual form is:

$$\text{CCR} = \text{cost before compression} / \text{cost after compression}$$

subject to preserved correctness and scope.

12.6 Exploration Diversity Index (EDI)

Measure diversity of active hypotheses, methods, role types, or search branches under comparable resources.

The goal is not maximal diversity, but sufficient diversity to avoid premature convergence.

12.7 Institutional Improvement Competence (IIC)

Define organizational improvement competence as:

$$\text{IIC} = \text{validated downstream gain from institutional proposals} / \text{proposal} + \text{evaluation cost}.$$

O4 requires evidence that IIC itself can improve across generations.

12.8 Topology Adaptation Gain (TAG)

Measure whether reconfiguration of communication or roles improves performance on changed task distributions relative to a frozen topology.

12.9 Turnover Resilience (TR)

Measure retained organizational performance after removal or replacement of members, including strong members.

12.10 False-Knowledge Propagation Rate (FKPR)

Measure how often unsupported or incorrect local claims become organizationally promoted and influence downstream work.

12.11 Organizational Frontier Expansion Index (O-FEI)

Measure the growth of problem classes on which the organization can make reliable progress under a fixed or normalized resource budget.

12.12 Governance Integrity

Track unauthorized operations, conflicts of authority, evaluator contamination, provenance breaks, and policy drift separately from performance.

13. A Falsifiable Experimental Program

Experiment 1 — Coordination baseline

Compare one strong agent, several independent agents, a fixed-workflow multi-agent system, and a persistent organization.

Question: Does organization produce genuine synergy after cost is included?

Experiment 2 — Organizational memory under member turnover

Train an organization on repeated tasks, replace a fraction of members, and measure retention.

Pass condition: critical lessons and procedures remain available without requiring the departed members.

Experiment 3 — Adaptive role allocation

Create nonstationary task distributions in which member strengths change over time.

Question: Can the organization learn competence maps and reassign roles faster than a fixed router?

Experiment 4 — Expert utilization versus consensus trap

Construct teams with one demonstrably superior expert and several weaker members.

Question: Can the organization use expert evidence without blindly centralizing all future decisions?

This directly addresses the anti-synergy observed in recent LLM-team studies [10].

Experiment 5 — Exploration-exploitation balance

Compare an organization that greedily copies the current best strategy with one that preserves protected exploratory branches.

Endpoints: short-term performance, long-term best solution, diversity, and recovery from distribution shift.

Experiment 6 — Knowledge diffusion and cognitive compression

Require a high-level member to discover a difficult method, then measure whether the organization can compile it into a tool usable by cheaper members.

Pass condition: later tasks retain correctness while substantially reducing high-level cognitive cost.

Experiment 7 — Institutional plasticity

Introduce a task distribution for which the initial communication topology is deliberately suboptimal.

Question: Can the organization propose, test, and promote a superior structure?

Experiment 8 — Recursive organizational improvement

Run multiple generations:

$O_0 \rightarrow O_1 \rightarrow O_2 \rightarrow \dots$

where each generation can improve the mechanism that proposes and evaluates organizational changes.

Endpoints: IIC, regression rate, improvement yield, and evaluation cost.

Experiment 9 — Autonomous collective discovery

Use hidden-law environments requiring hypothesis generation, division of experimental work, evidence integration, and new rule discovery.

Question: Can the organization autonomously create a research agenda rather than merely execute assigned subproblems?

Experiment 10 — Adversarial misinformation and epistemic resilience

Insert faulty, overconfident, or strategically misleading members.

Endpoints: FKPR, recovery time, expert leverage, provenance integrity, and downstream contamination.

Experiment 11 — Turnover and shock resilience

Remove critical members, corrupt part of shared memory, or change resource availability.

Question: Does the organization detect the loss, reconstruct capability, and preserve mission continuity?

Experiment 12 — Open-ended frontier expansion

Use a long-running environment in which new discoveries unlock new task families and new organizational forms.

O Ω evidence requires: repeated frontier expansion caused by organizationally generated tools or institutional changes, not merely more compute or externally supplied tasks.

14. Relationship to Prior Work

14.1 Collective intelligence

Human-group research motivates the idea that group performance can possess structure not captured by the strongest individual alone [1]. The exact independence of collective intelligence from individual intelligence remains contested, so O Ω does not assume a universal latent “c factor.” Instead it defines organizational intelligence operationally through persistence, learning, reconfiguration, synergy, and frontier expansion.

14.2 Organizational learning

Argyris and Schön’s distinction between single-loop and double-loop learning provides a conceptual ancestor for O2 and O3, while later work on deutero-learning or learning-to-learn is closely related to O4 [2,3]. O Ω extends this logic into artificial organizations whose institutional mechanisms can be explicitly versioned and experimentally evaluated.

14.3 Exploration and exploitation

March’s analysis remains central because an organization that always copies the current winner can become increasingly efficient while destroying the diversity needed for future discovery [4]. O Ω therefore treats exploration diversity as a protected organizational resource rather than accidental inefficiency.

14.4 Organizational memory

Walsh and Ungson’s treatment of organizational memory as acquisition, retention, and retrieval maps naturally onto persistent artificial organizations [5]. O Ω adds evidence provenance, machine-readable procedures, member turnover tests, and active memory governance.

14.5 Organizational knowledge creation

Nonaka’s theory emphasizes that knowledge may originate in individuals while organizations articulate and amplify it [6]. O Ω operationalizes this as local-to-global promotion followed by tool compilation and distributed adoption.

14.6 LLM multi-agent systems

Recent surveys of LLM-based multi-agent systems emphasize architecture, memory, planning, communication, and scalability [11]. These are necessary substrates but do not by themselves establish O3–O Ω because a fixed multi-agent architecture may remain organizationally static.

14.7 Group-Evolving Agents

Group-Evolving Agents treat a group rather than an isolated branch as the evolutionary unit and use experience sharing to convert exploratory diversity into sustained progress [7]. This is close to the O Ω idea of organizational evolution, though O Ω additionally requires persistent organizational identity, institutional lineage, organizational memory, governance, and open-ended frontier expansion beyond a benchmarked evolutionary search.

14.8 CORAL

CORAL explicitly targets autonomous multi-agent evolution for open-ended discovery using persistent memory, asynchronous agents, exploration, reflection, and collaboration [8]. It provides strong evidence that open-ended

organizational search can outperform fixed evolutionary heuristics. O Ω generalizes the target from a discovery framework to a developmental class of organization.

14.9 Evolving institutions

Recent work translating historical governance structures into executable multi-agent architectures shows that governance topology can shift collective performance dramatically and that the best organization may depend on model capability and task structure [9]. This supports a core O Ω claim: no single fixed organization is universally optimal; the ability to reselect and reconfigure institutional form is itself an intelligence capability.

14.10 The expert-utilization problem

The finding that self-organizing LLM teams may fail to match their expert member demonstrates that coordination can destroy rather than create intelligence [10]. O Ω therefore treats expert leverage, dissent preservation, and evidence weighting as first-class organizational functions.

14.11 Relationship to I Ω

I Ω studies one persistent cognitive lineage that discovers knowledge and converts it into better cognition. O Ω studies a persistent **collective** lineage in which knowledge, tools, roles, and institutions can be distributed across members and survive membership change.

The two loops can couple:

Individual evolution ↔ **organizational evolution.**

An improved individual can contribute a new organizational method. An improved organization can provide individuals with better memory, evidence, tools, tasks, collaborators, and evaluation, enabling further individual development.

15. What O Ω Is Not

15.1 Not simply many agents

Parallelism is not organizational intelligence.

15.2 Not merely a router

A fixed dispatcher may coordinate perfectly while never learning.

15.3 Not merely shared memory

Shared memory without validation can accelerate error propagation.

15.4 Not merely self-organization

Spontaneous interaction does not imply useful adaptation; unconstrained teams can show anti-synergy.

15.5 Not merely a swarm

Swarm intelligence can be powerful, but O Ω requires persistent organizational memory, explicit knowledge integration, institutional lineage, and open-ended organizational learning.

15.6 Not simply an organization containing I Ω individuals

O-level is not the maximum I-level of the membership.

15.7 Not a civilization scale

O Ω concerns one bounded organization. A civilization may contain many O Ω organizations plus markets, states, communities, humans, machines, and infrastructures. Civilization-scale intelligence is a further level of analysis.

15.8 Not permanent structural change

An organization that continuously rewrites itself without stable gains is unstable, not open-endedly intelligent.

16. Failure Modes and Risks

16.1 Consensus trap

Members converge on compromise even when one member has superior evidence.

16.2 Epistemic monoculture

The organization over-standardizes models, tools, or beliefs and loses exploratory diversity.

16.3 Organizational amnesia

Knowledge remains trapped in individuals or ephemeral conversations and disappears after turnover.

16.4 Institutional lock-in

A once-successful topology becomes too entrenched to change when tasks shift.

16.5 Organizational thrashing

The organization changes structure too frequently to accumulate reliable evidence about any configuration.

16.6 Metric capture

Institutional changes optimize measured performance while degrading unmeasured capabilities such as exploration, calibration, or resilience.

16.7 False-knowledge cascades

A high-status or high-throughput member contaminates shared memory faster than verification can respond.

16.8 Hidden single-member dependence

The organization appears intelligent but actually relies on one expert whose removal collapses performance.

16.9 Coordination overhead

Communication, debate, and verification consume more resources than the collective gain they produce.

16.10 Authority drift

A change framed as organizational optimization silently expands who may authorize consequential actions. Capability evolution and authority evolution must therefore remain separable.

16.11 Faction and goal fragmentation

Persistent members can develop incompatible local objectives, creating internal competition that is not visible in aggregate task metrics.

16.12 Homogeneous correlated failure

Using the same model, prompt, data, and reasoning pattern across every member can create the illusion of independent consensus while reproducing one shared mistake.

17. A Staged Roadmap Toward O Ω

Stage 0 — Coordinated multi-agent system

Implement explicit roles, bounded communication, and task decomposition. Measure whether the system provides synergy at all.

Stage 1 — Persistent organization

Add shared organizational memory, evidence lineage, current commitments, role state, and restart continuity.

Stage 2 — Adaptive organization

Use outcome history to update expertise maps, routing, procedures, and shared knowledge. Demonstrate improvement on held-out future tasks.

Stage 3 — Organizational self-improvement

Permit candidate modifications to topology, role structure, memory policy, and workflows under versioned evaluation.

Stage 4 — Recursive organizational meta-improvement

Allow the mechanism that proposes and evaluates institutional changes to improve under independent testing.

Stage 5 — Autonomous discovery organization

Add structured unknowns, research-agenda generation, hypothesis portfolios, evidence collection, replication, and knowledge promotion.

Stage 6 — Cognitive compression and institutional invention

Demonstrate that discoveries become shared tools, lower-cost procedures, new roles, and new organizational structures.

Stage 7 — Dynamic cognitive ecology

Support member creation, specialization, retirement, topology adaptation, cross-domain transfer, and protected exploratory diversity.

Stage Ω — Sustained open-ended organizational evidence

Claim O Ω only after multiple generations show that organizationally generated discoveries and institutional changes repeatedly expand the collective frontier, while memory, provenance, turnover resilience, governance, and cost-adjusted performance remain viable.

18. A Reference Realization Pattern for AI Agent Organizations

A practical artificial organization can separate persistent individual brains from organizational infrastructure.

One candidate pattern is:

persistent role agents → **shared evidence ledger** → **organizational memory** → **task/role router** → **verifier network** → **institution foundry** → **evaluator/promoter** → **tool/procedure registry**.

Persistent individual agents maintain private autobiographical state and local expertise. Organizational memory stores shared validated knowledge and institutional history. A manager or coordination layer does not automatically become the “smartest” agent; its role is to maintain scope, routing, shared state, and cross-member communication. Specialized research agents may themselves internally coordinate sub-agents, creating nested organizations.

This pattern is compatible with AI4Science-style systems in which persistent role agents coordinate executors and research agents can be group agents with sub-agents. Such systems provide a useful engineering substrate for testing O-level transitions, but architecture alone does not establish O Ω . The required evidence remains longitudinal: organizational learning, institutional improvement, recursive improvement competence, autonomous discovery, cognitive compression, and frontier expansion.

19. Discussion

19.1 The organization is a cognitive object in its own right

The central claim of this paper is ontological as much as architectural. Once memory, roles, knowledge, procedures, and improvement history persist beyond individual members, the organization becomes a legitimate cognitive unit

of analysis. It can remember things no current member personally experienced. It can preserve procedures none of its current members invented. It can create new roles in response to knowledge generated by earlier members. It can therefore possess a developmental history that is not reducible to any one individual history.

19.2 OΩ may appear before IΩ

It is possible that open-ended organizational intelligence is easier to realize than open-ended individual intelligence. Evolution, diversity, replication, independent evaluation, and specialization are naturally distributed across populations. An organization can externalize cognition into institutions and shared memory rather than requiring one individual to contain every function.

Thus an advanced artificial organization composed of I2–I5 members might exhibit stronger open-ended development than any one member.

19.3 IΩ may also destabilize weak organizations

The opposite possibility is equally important. A very powerful individual placed inside an O1 organization can create dependency, overload shared memory, dominate consensus, or bypass organizational learning. As individual intelligence rises, organizational design becomes more—not less—important.

19.4 The most advanced organization may look simpler at the execution edge

If cognitive compression works, frontier discoveries should produce increasingly simple interfaces for routine work. The most sophisticated organization may therefore contain a large surface of inexpensive, low-level execution supported by a much smaller deep layer of discovery and institutional improvement.

19.5 Open-endedness is a property of lineage, not a snapshot

No static benchmark can prove OΩ. A snapshot can show capability; OΩ is a claim about developmental capacity across changing environments and generations.

The evidence must therefore be longitudinal:

O_0 → O_1 → O_2 → ...

with traceable organizational changes, causal hypotheses, measured outcomes, preserved provenance, and repeated frontier expansion.

20. Conclusion

This paper proposes **OΩ — Open-Ended Evolving Organization** as the highest developmental class of organizational intelligence in a two-axis framework that separates individual and collective evolution.

The defining feature of OΩ is not agent count, model size, communication volume, or the presence of a powerful manager. It is a persistent collective lineage capable of converting distributed experience into evidence-bearing organizational knowledge; converting knowledge into better shared procedures, tools, roles, and institutions; improving the process that generates future organizational improvements; autonomously expanding the knowledge frontier; and preserving enough diversity, memory, resilience, and governance to continue that process across changing members, tasks, and environments.

The compact formulation is:

Distributed experience becomes organizational knowledge. Organizational knowledge becomes institutions and tools. Institutions and tools improve collective intelligence. Improved collective intelligence reaches a deeper frontier. The deeper frontier generates new experience, and the organization can improve the process that drives the cycle itself.

In symbolic form:

Experience → Evidence → K^O → {Tools, Roles, Topology, Institutions} → Better Organization → Deeper Frontier → Experience'.

The framework also explains why all individual intelligence levels can coexist inside the highest organization. High-level members discover and redesign; middle levels adapt and translate; low-level members execute mature procedures efficiently. Through cognitive compression, the output of expensive intelligence becomes the infrastructure of cheap intelligence. Through upward epistemic flow, local anomalies can eventually change the organization itself.

The strongest organizational intelligence is therefore not the organization that contains the smartest member. It is the organization that can **continually transform distributed intelligence into shared capability, and shared capability into new forms of collective intelligence that make previously unreachable problems reachable.**

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